Capstone

December 4 launch · December 7 studio · December 9 gallery

Last updated: August 18, 2026.

Dates are on the schedule. Full write-up instructions will be added here. Use this packet to prepare; there is no quiz on SVD.

Octave syntax: Lachniet Ch. 5 (SVD) and Ch. 7 (image compression). One PDF on Resources; this handout is the assigned work.

Computing an SVD does **not** automatically make every later calculation faster. The factorization itself can be expensive. Compression or faster multiplication is valuable when the matrix is well approximated at low rank **and** the representation will be stored, transmitted, or reused enough times to justify that initial cost. Image examples are instructional; adaptive SVD representations can themselves be expensive.

Practice packet

About 14 prompts. Source key: Assessments.

- S 7.1 #1, #2, #3, #5, #6
- L Ch. 5 Ex. 7: small SVD
- L Ch. 5 Ex. 8: pseudoinverse/least-squares connection
- L Ch. 7 Project 7.1, Problems 1–3: image compression
- C-Cap.1: Derive full storage mn.
- C-Cap.2: Derive rank-k storage $k(m+n+1)$.
- C-Cap.3: Plot relative reconstruction error versus k.
- C-Cap.4: Compare full and factorized matrix-vector multiplication times and estimate when the initial SVD cost pays for itself.

Learning objectives (launch day)

See Lecture 40.